AUTOCAD AND SOLIDWORKS
FOR
MECHANICAL ENGINEERS

A CERTIFICATE COURSE CONDUCTED

BY

THE SURE Trust

Skill Upgradation for Rural-youth Empowerment – Trust

https://suretrustforruralyouth.com/

Course Training Attended

By

Riddesh Ravindra Hirejagner

2025-2026

DECLARATION

This is to certify that Mr. **Riddesh Ravindra Hirejagner** has successfully completed the 6 months training given in "**AutoCAD and Solid Works for Mechanical Engineering**" by SURE Trust during the period from November 2025 to April 2026

Trainer:

Mr. MAIRALA CHINNARAO

Prof.Ch. Radha Kumari

Executive Director & Founder,

SURETrust

Mrs. Vandana Nagesh

Director & Co-Founder,

SURE Trust

TABLE OF CONTENTS

About the SURE Trust

Introduction to the SURE Trust

The SURE Trust is born to enhance the employability of educated unemployed rural youth. It is observed that there is a wide gap between the skills acquired by students from the academic institutions and the skills required by the industry to employ them. Employability enhancement is done through giving one on one training in emerging technologies, completely through online mode . The mission of the SURE Trust is to bridge the gap between the skills acquired and the skills required by training them in the most emerging technologies such as Autocad & Solidworks, Python Program, Machine Learning (ML), Deep Learning (DL), Data Science & Data Analytics, Block chain Technology, Robotic Process Automation (RPA), and Project Management and twenty other essential and high in demand Courses ,that will enhance their employability. After completion of four months training in the course, the trainees will get live projects from industries as internship activity to get experience in applying to real time situation what they have learnt during the course. These projects will give them hands on experience which is much sought after by the prospective industry employing them. Currently students from all over India are enrolling for various courses offered by the SURE Trust. The SURE Trust offers every course free of cost with no financial burden of any kind to students. This initiative is purely a service-oriented one aiming to guide the rural youth who are educated but unemployed due to lack of upgradation in their skill sets. The birth of SURE Trust is a God given boon to rural youth who could reach great heights either in employment or in entrepreneurship once they receive the training offered followed by the company internship. Many companies are coming forward to join their hands with us by offering internship projects to hand hold and lead the rural youth in their career settlement.

Vision of the SURE Trust

The vision of the SURE Trust is to enhance the employability of educated unemployed youth, particularly living in rural areas, through skill upgradation, with no cost to the students.

Mission of the SURE Trust

The mission is to bridge the gap between the skills acquired in the academic institutions and the skills required in industries as a pre-condition for employment.

Functioning of the SURE Trust

There are three dedicated, committed, and hard-working women on the board of management of the SURE Trust who will look into the various administrative and other matters relating to the enrolment of students, organizing trainers, entering into agreements with companies for getting live projects to students as internship programs, and so on.

All the three women on the board are all the alumni from Sri Sathya Sai Institute of Higher Learning, Anantapur Campus, deemed to be a University. The women board is supported by five eminent advisories who are from different walks of life and have made outstanding mark in career in their respective fields. For more details about SURE Trust please visit the website www.suretrustforruralyouth.com.

2. Course Content

The SURE Trust conducts a five months training for every course on a uniform basis. A session spanning across one to one & half hour is taken by the trainers for every major course. Sessions are conducted to complete the predesigned course structure within the fixed time period. Course content is designed to suit the current requirement of the Industry and validated by the industry experts. The course content of all these courses is so dynamic that any changed condition noticed in the industry will automatically get reflected in the content of the respective course . As the course content is dynamic, the Following is the course content of the current course in Data Science and Data Analytics:

AUTOCAD & SOLIDWORKS FOR MECHANICAL ENGINEERS

Objective:

The objective of this course is to train mechanical engineering students in the basic design software.

Course content:

Module 1. Solidworks:
SolidWorks is a solid modeling computer-aided design (CAD) and computer-aided engineering (CAE) application published by Dassault Systems. The topics covered are:
- Introduction
- Sketch and Various Sketch Tools in SolidWorks
- 3D Part Modelling Using SolidWorks Features
- SolidWorks Assembly and Its different Features
- Sheet metal, Weldments and Mold Tool
- Basics of SolidWorks Rendering
- Drafting using SolidWorks Drawing

Module 2. Autocad:
AutoCAD is a widely used software application that allows engineers to draw error-free designs. As such, its functions and features must be understood properly by aspiring engineering students. The main topics covered are:
- Introduction to AutoCAD
- Getting started with toolsets generating design drawings with CAD
- Drafting styles and standards
- Annotation units and conversion methods
- Clear and error-free drawings

3.Conduct of the course

a) Modalities for the conduct of all the courses are fixed by the SURE Trust which are uniformly followed across the courses.

- Mode of Training --- Online
- Period of Training --- Four months
- Sessions per week --- 3 to 6
- Length of the session --- 1 to 2 hours
- Tests to be taken --- 2 per month
- Assignments --- 2 per month
- Last 15 days --- Final practice and preparing the course report

b) Student byelaws:

- Students enrolling for the courses under SURE Trust are strictly required to follow the following byelaws set for them.

Byelaws for students to become eligible for certificate at the end of the course

I. Minimum Attendance:

- Every student must put in a minimum of 85% attendance in attending the classes for getting the eligibility to receive the certificates.

II. Two written tests are to be taken in each month:

Since the objective of the certification program is to turn out well qualified students from the respective courses, minimum two written tests are to be taken in each month for each course to ensure that the students are pulled along the expected line of standard.

III. Assignment submissions:

Ten exercises constituting one assignment for every two to three new functions/topics taught, resulting in minimum seven such assignments are to be submitted during the four months period.

IV. Preparing the final course report in the prescribed format:

During the last fifteen days in the fourth month, students many be asked to consolidate and compile all the assignments submitted in a word document along with the other chapters which will constitute a course report for each student. This report will be the unique contribution a student carries from the trust to show case the rigorous training he/she received during the four months period. Besides the report will stand as a testimony for the detailed learning a student has acquired in the chosen area. This will facilitate the industry in handpicking the required student for the job.

V. External Viva-voce:

Every student has to successfully clear the external viva-voce arranged in their respective course.

VI. KYC norms:

Each student wishing to enroll for the course must submit a written letter saying that he/she will not drop from the course until its completion, which will also be signed by father / mother besides the student himself / herself.

VII. Attend the full class:

All the students are expected to attend each class for full duration. Some students are observed moving out of classes after logging in which does not go well with the learning objective of students.

VIII. Ensure discipline in the group:

All the students are advised strictly to follow group etiquette and restrain from posting in the group any unethical messages or teasing messages or personal interactive messages. This group is purely created for academic purpose and hence only academic interactions should go on.

AutoCAD & SolidWorks

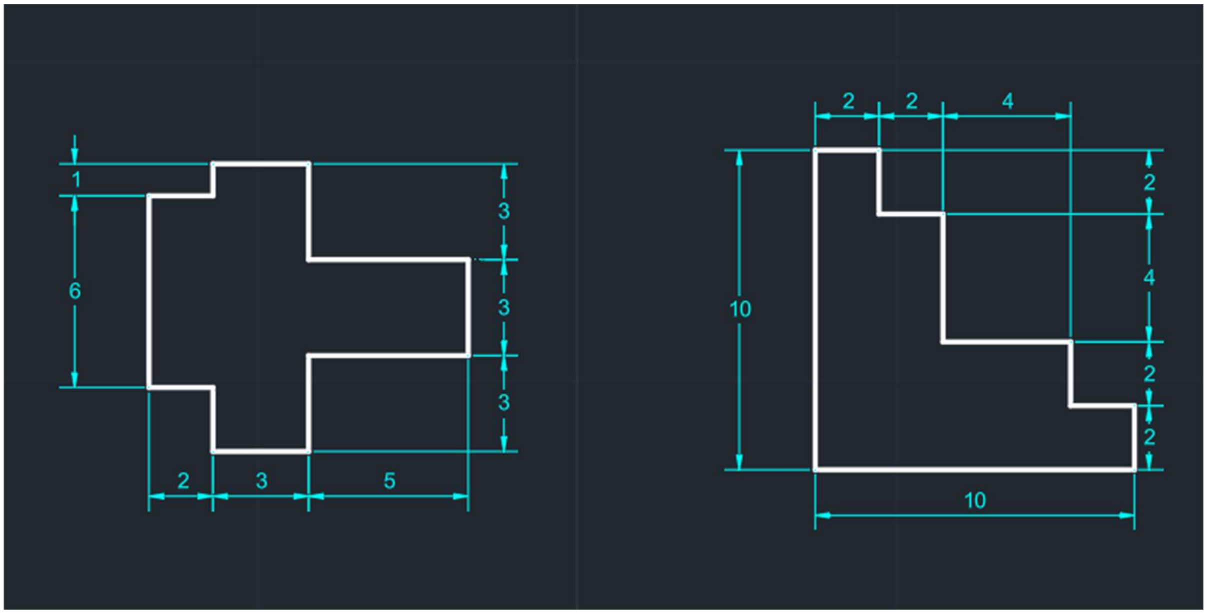
The course is taught to the students as per the syllabus prescribed and as per the course modalities keeping in mind the bylaws set for them. Periodical tests are conducted to assess the understanding of the students in the course. Assignments are given to ensure that students gain versality in the designing. The assignments of the student are given below. These assignments which are done with high creativity and out of box thinking not only reflect the student's innovativeness but also constitute solved exercises in the Various designs for the fresh learners to practice and gain confidence. The assignments are listed below in the order of their conducting during the course.

AutoCAD is a CAD program mainly used for architectural and construction designs, civil and structural engineering. On the other hand, SolidWorks is mainly used for mechanical parts and assembly design, technology designing, and automotive and aerospace designing.

Although AutoCAD and SolidWorks are computer-aided design software, their features differ greatly. Both have tools for 3D drawing but different aims. In AutoCAD, an object's projected view can be drawn, but in SolidWorks, a sketch is created, which can be further made into a 3D model. 3D modeling features are more dominant in SolidWorks as compared to AutoCAD. In AutoCAD, the toolset is pretty much hidden.

<u>ASSIGNMENTS</u>

Assignment 1 :



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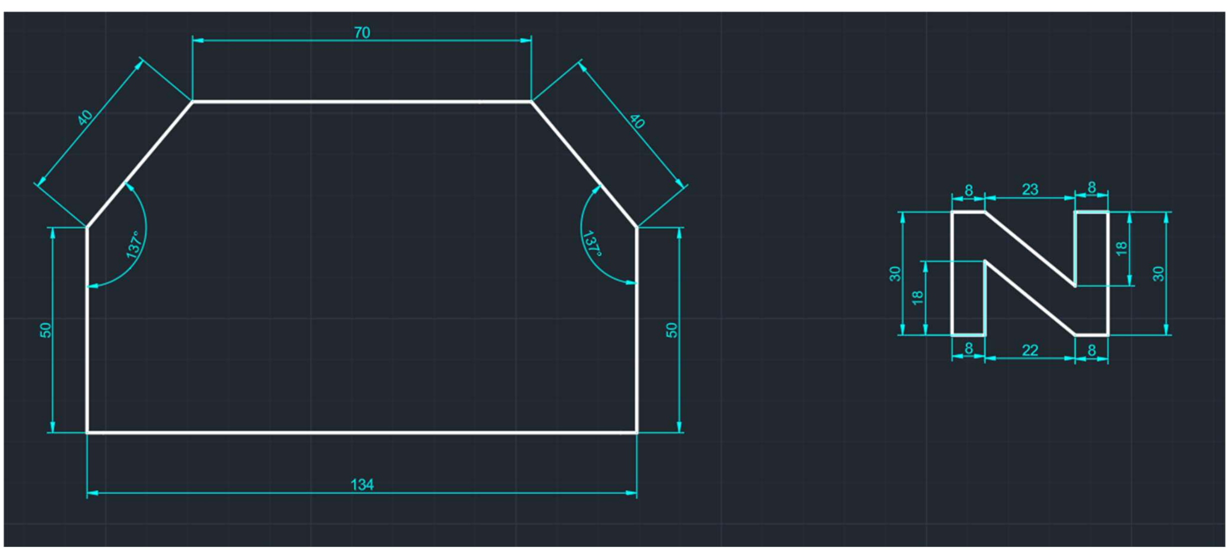

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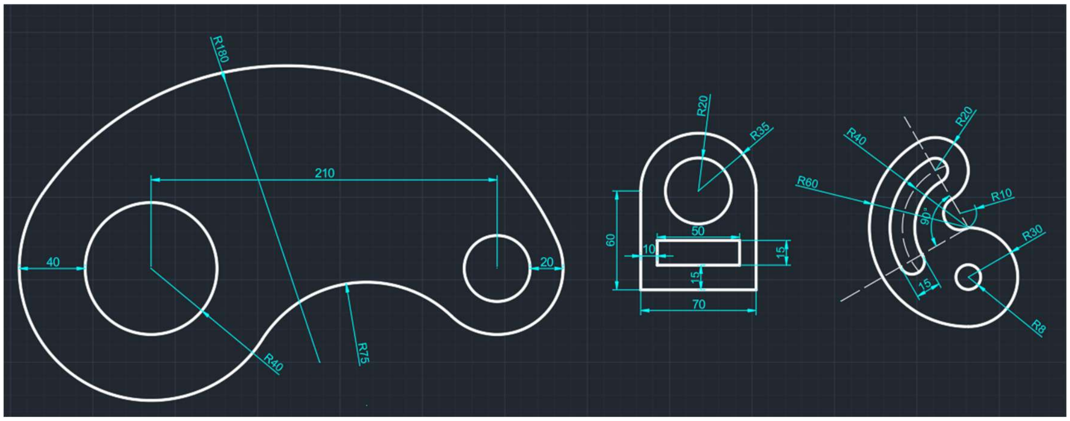

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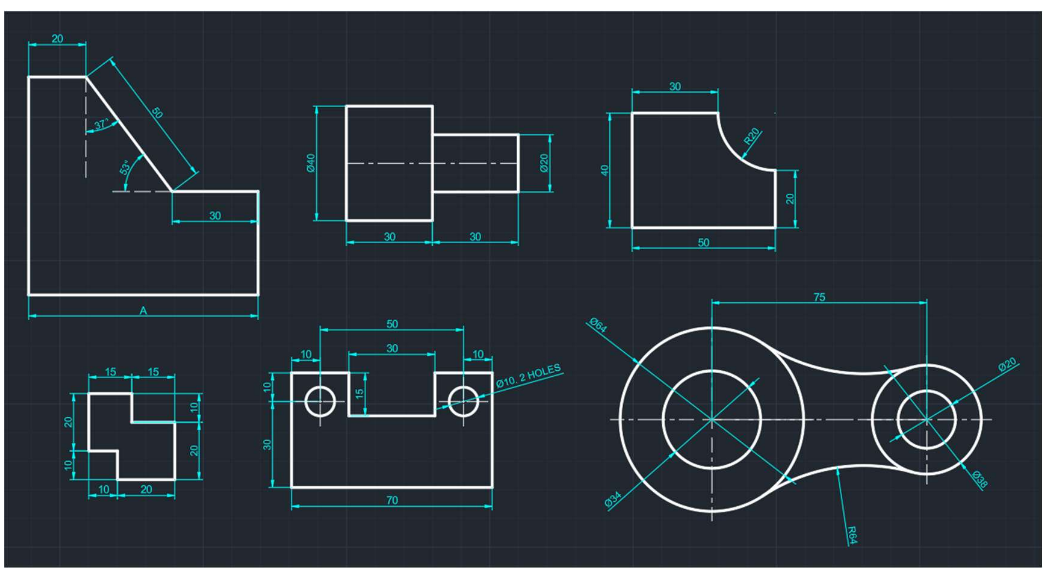

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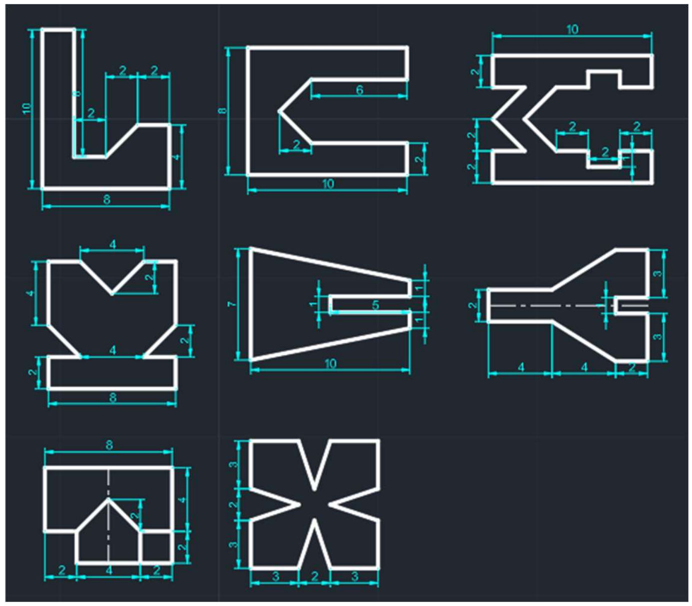

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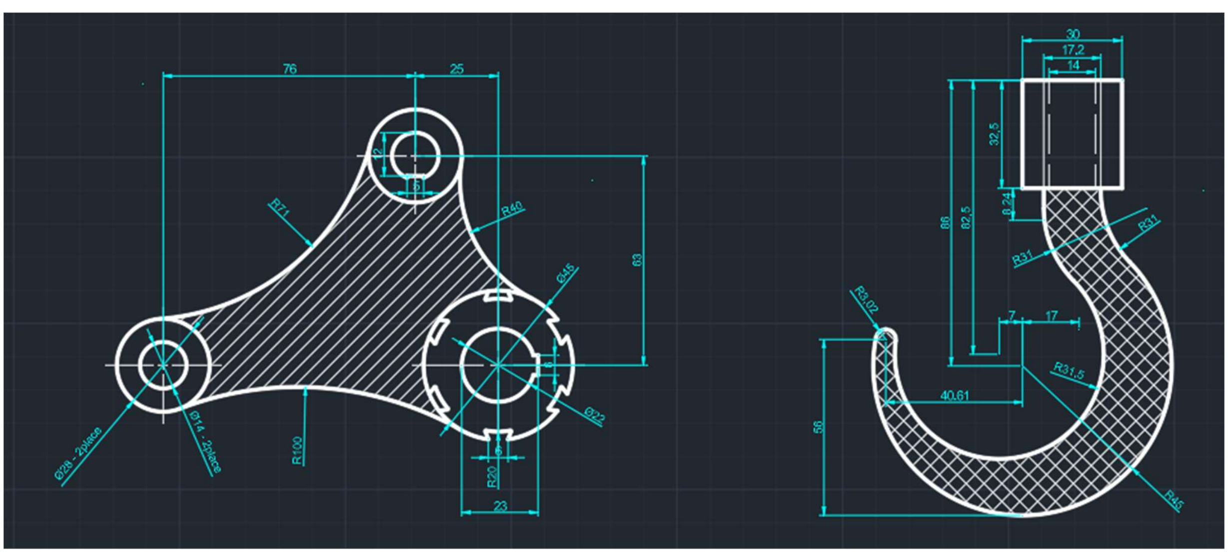

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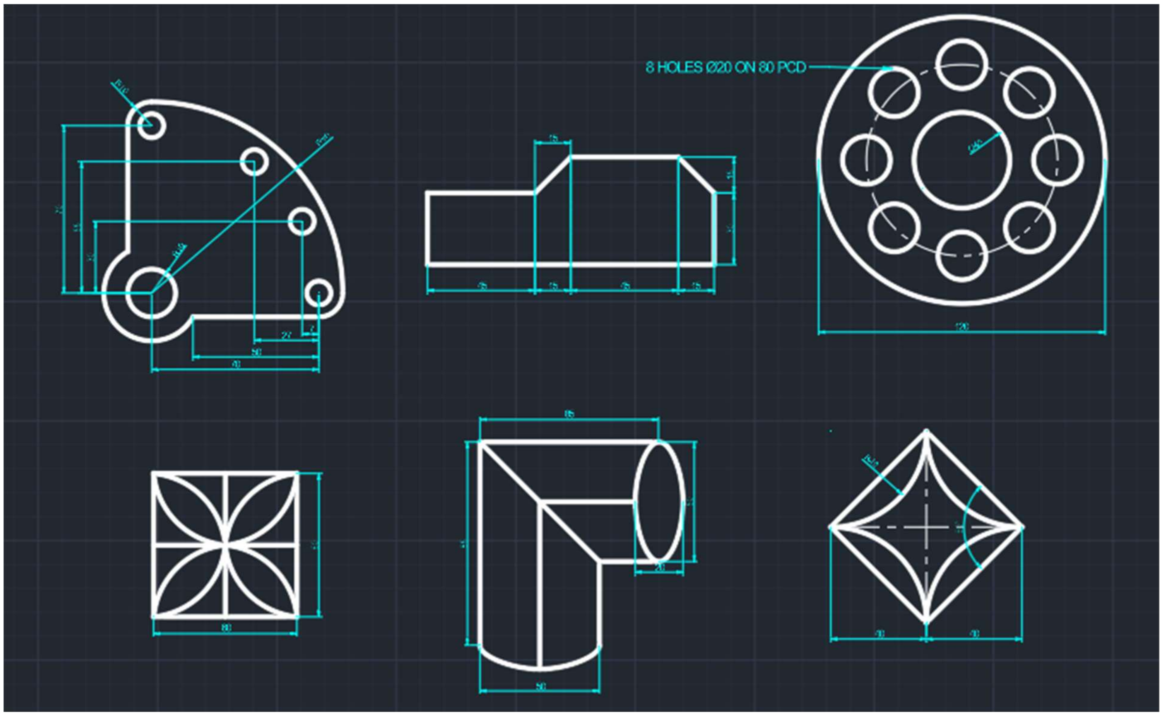

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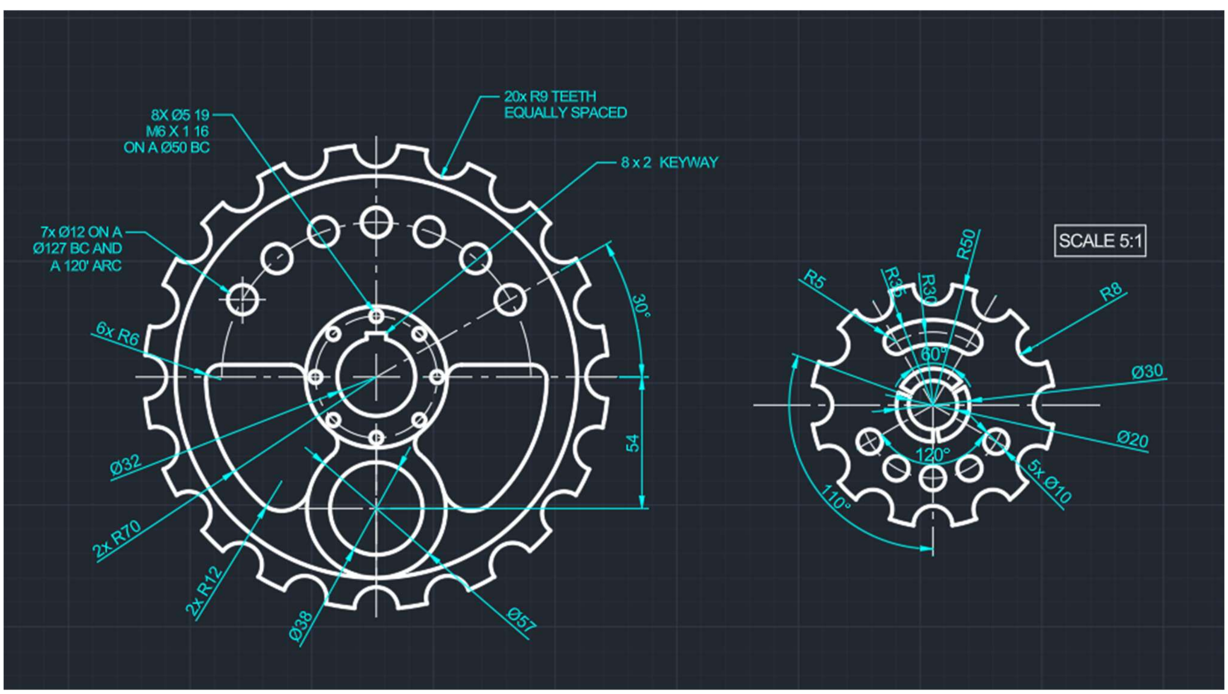

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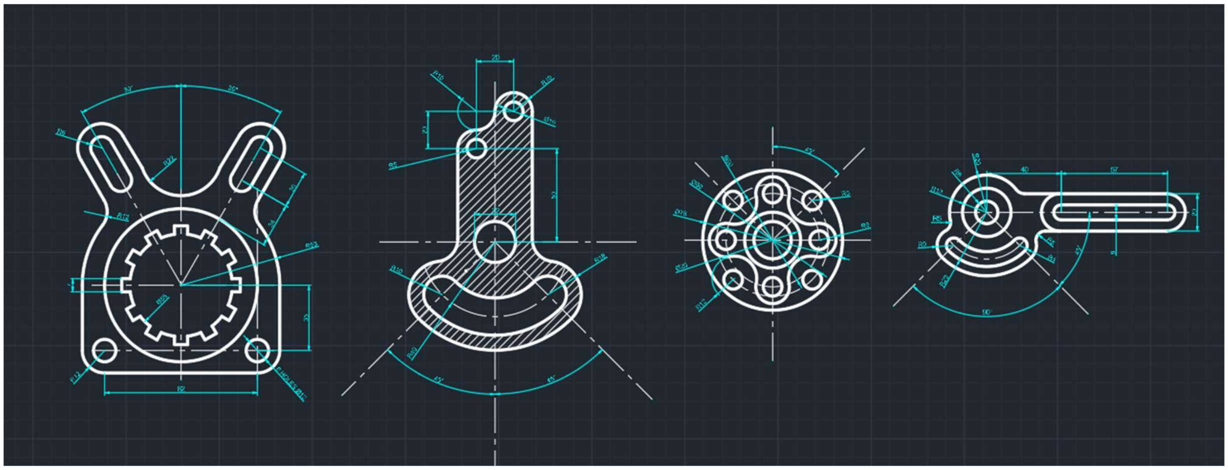

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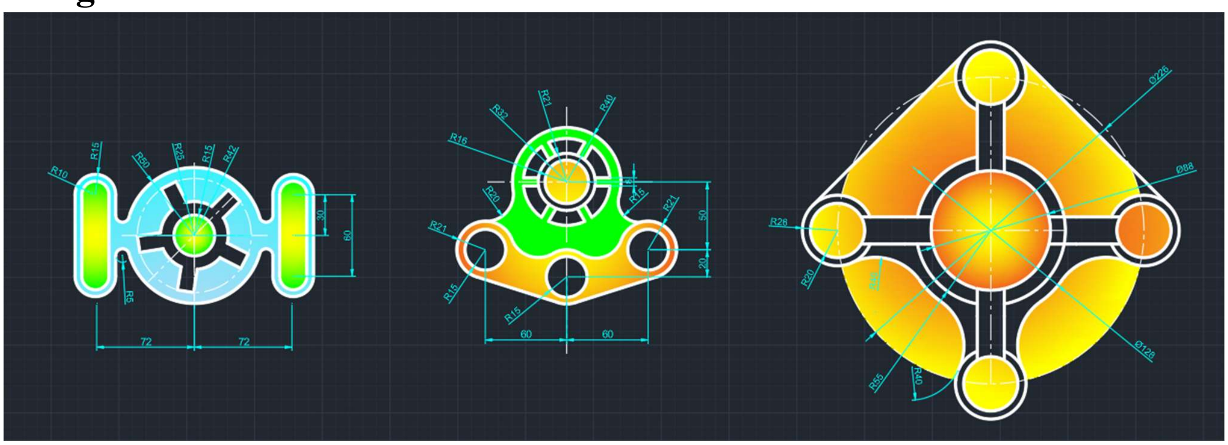

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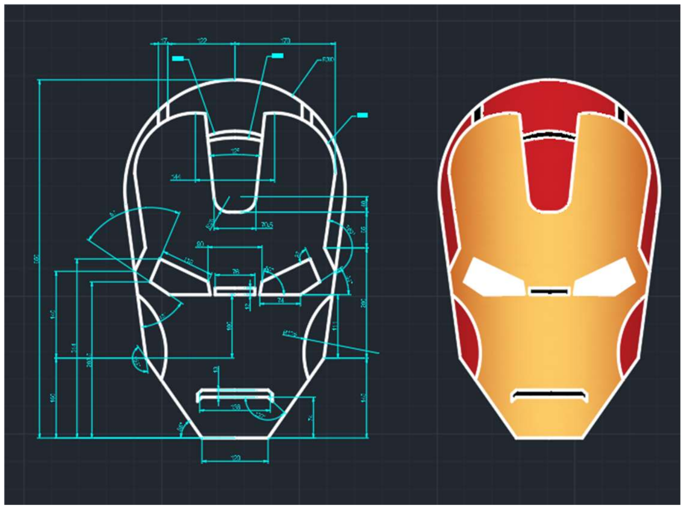
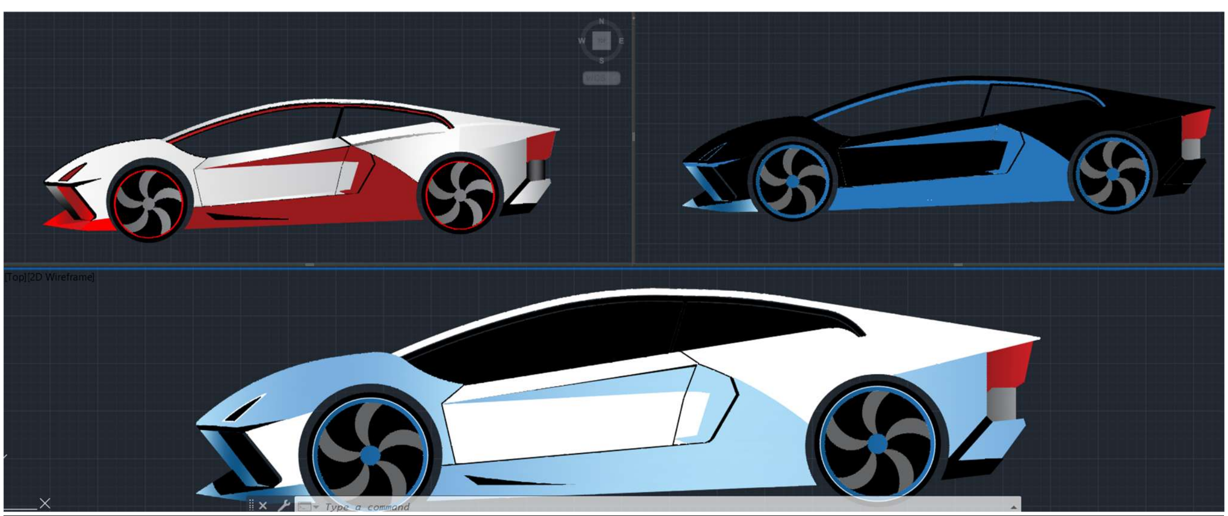
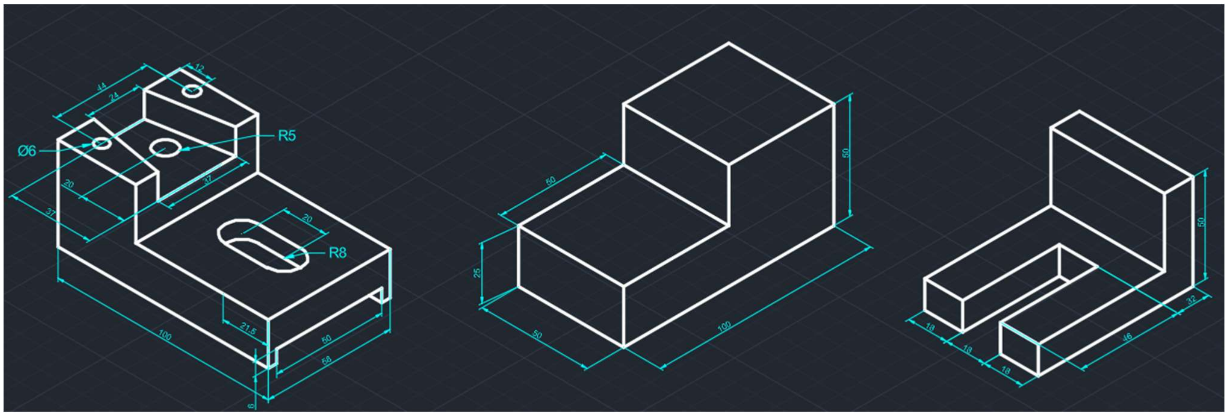


Miniproject: Sports Car Design



Iso Drawing

Assignment 1:



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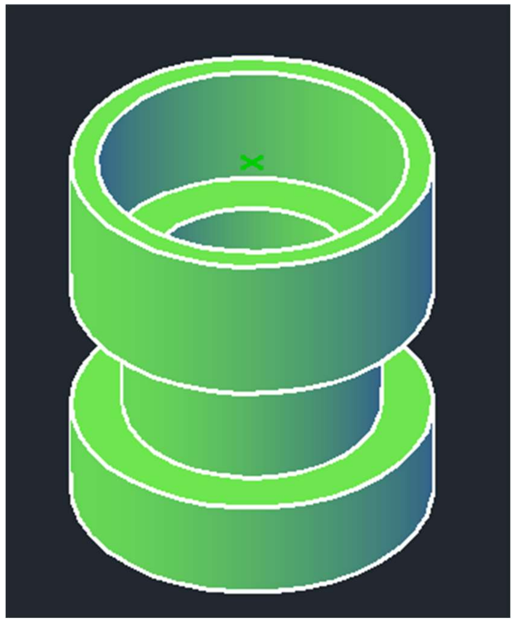

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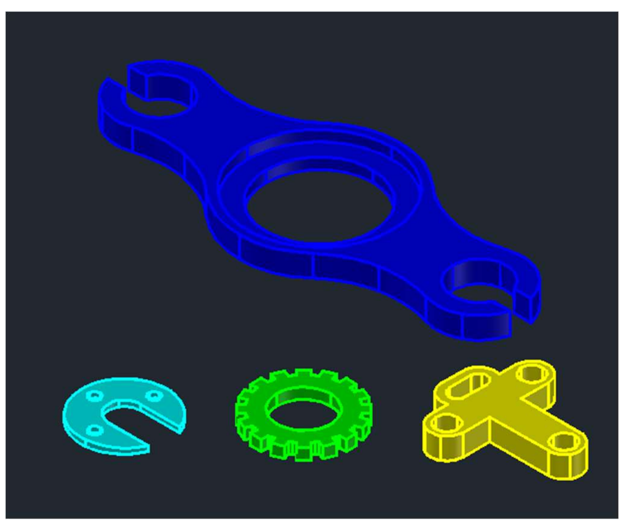
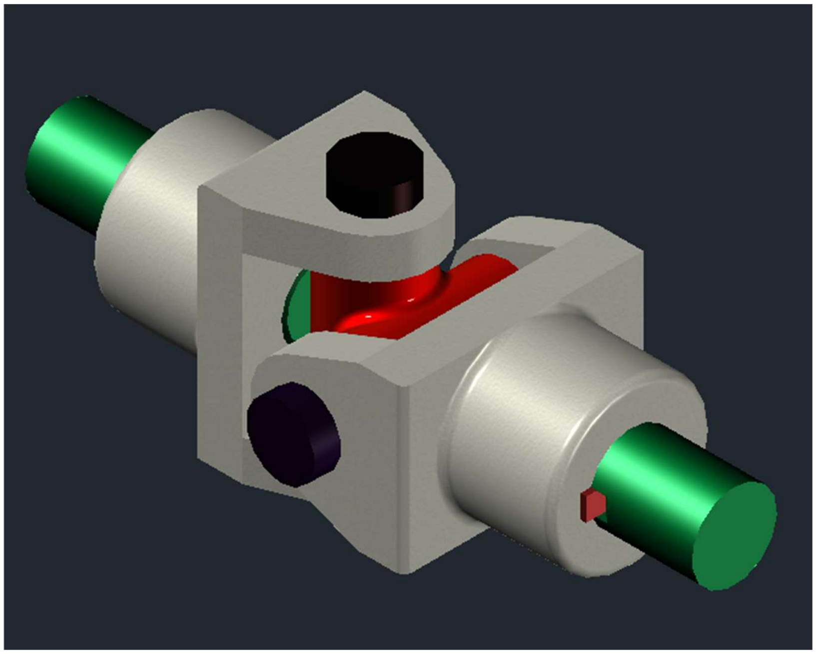


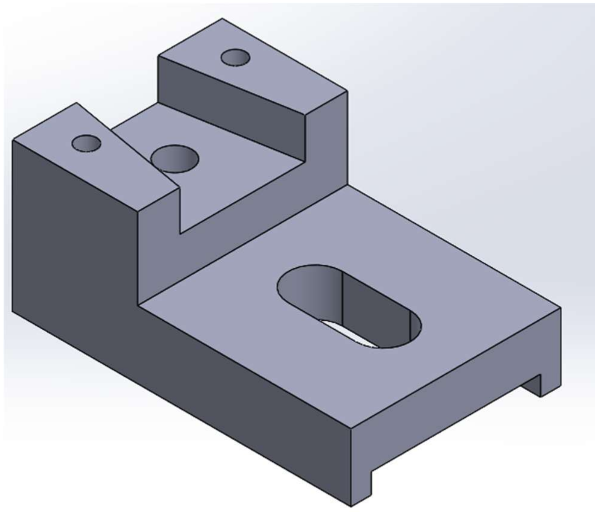
Major Project:



SOLIDWORKS

Part Modelling

Assignment 1:



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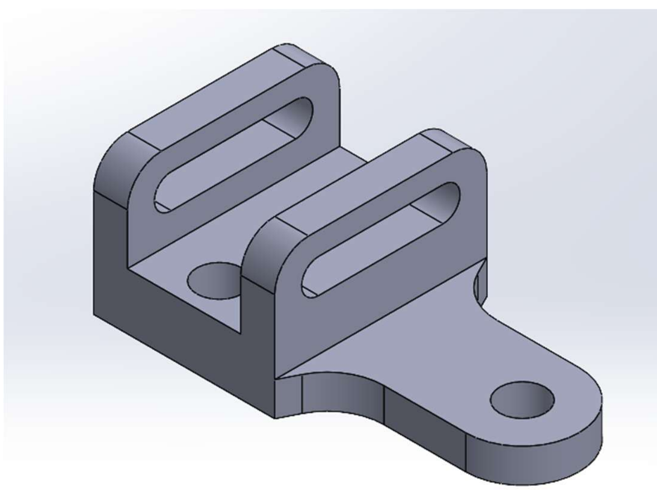

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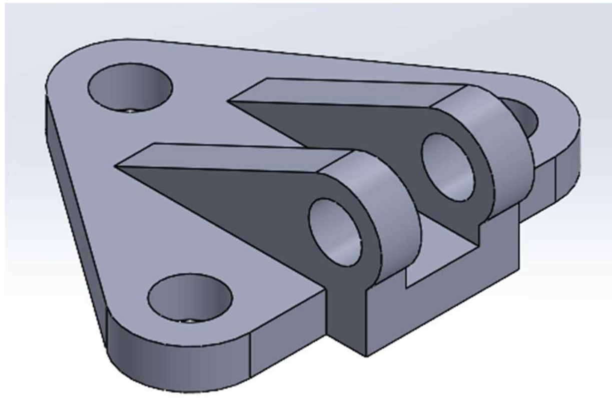

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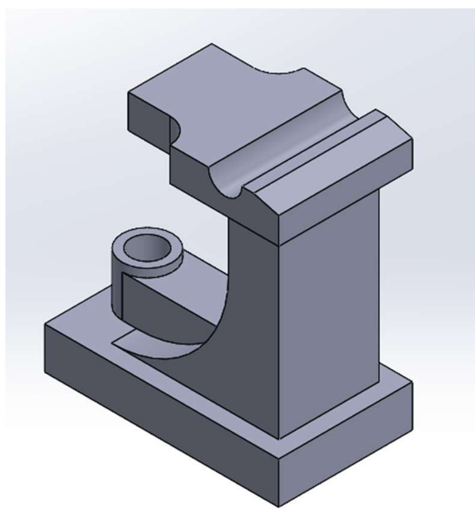

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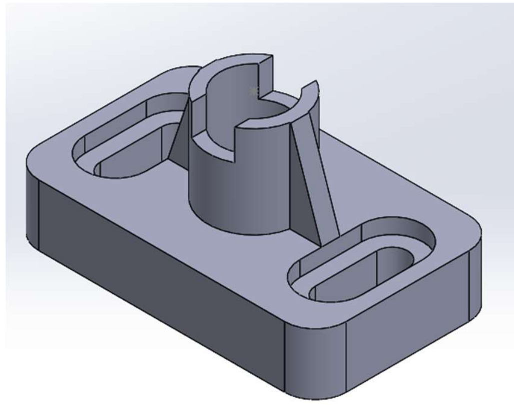

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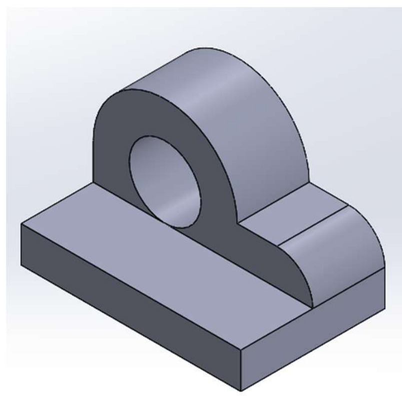

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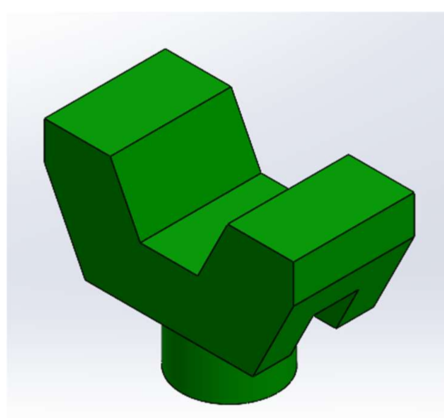

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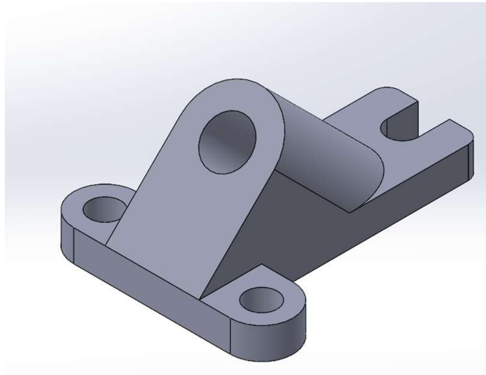

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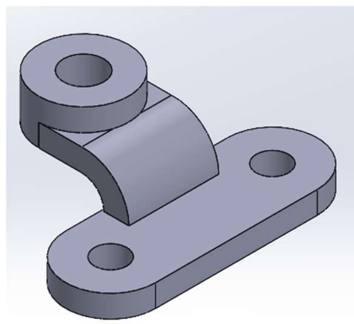

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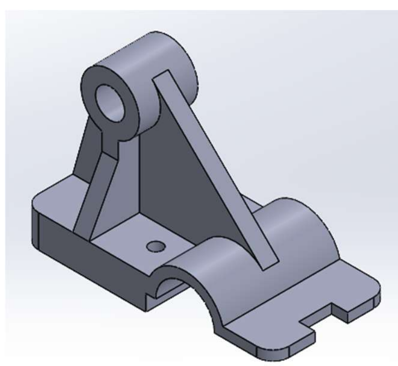

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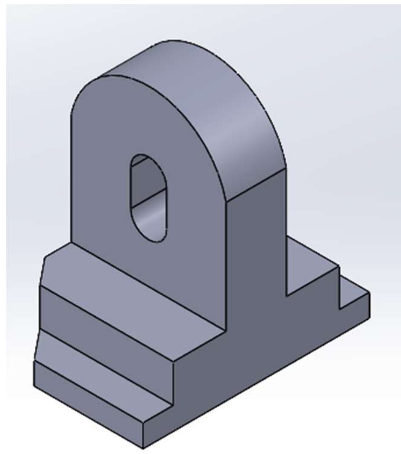

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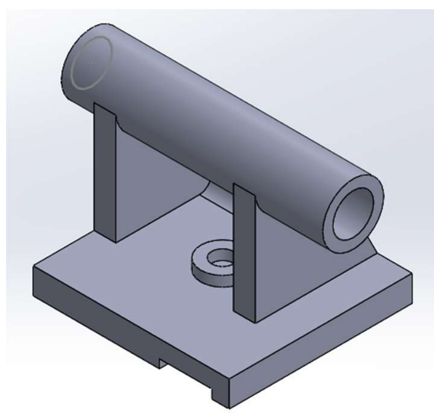

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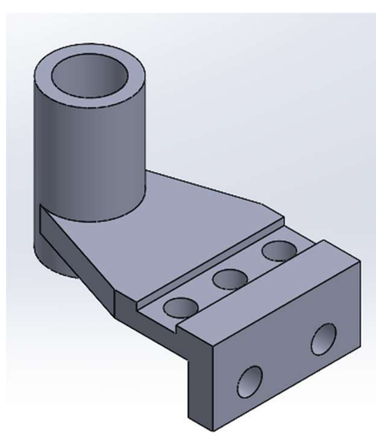

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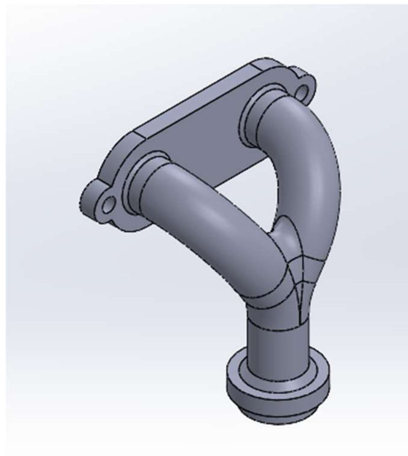

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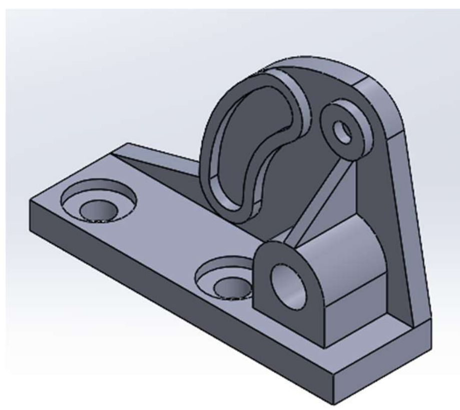

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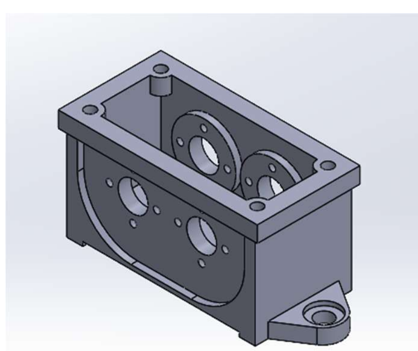

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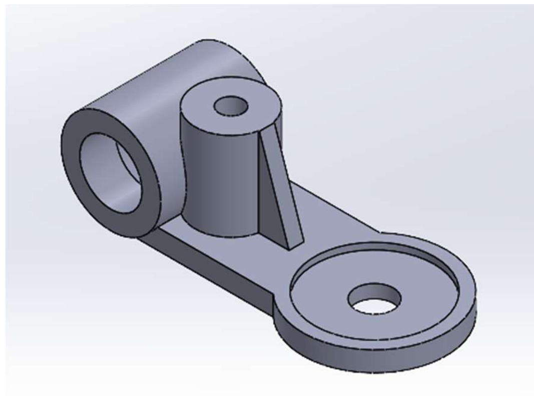

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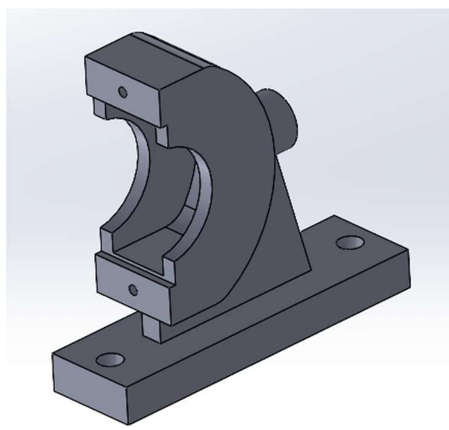

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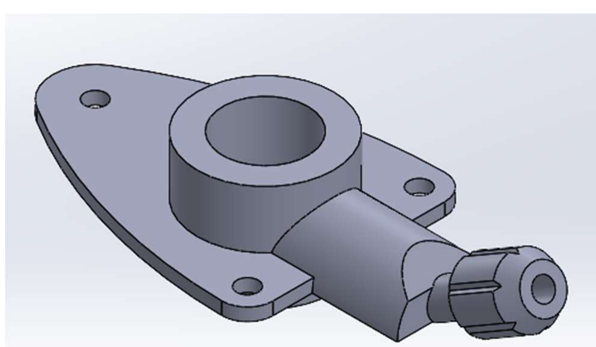

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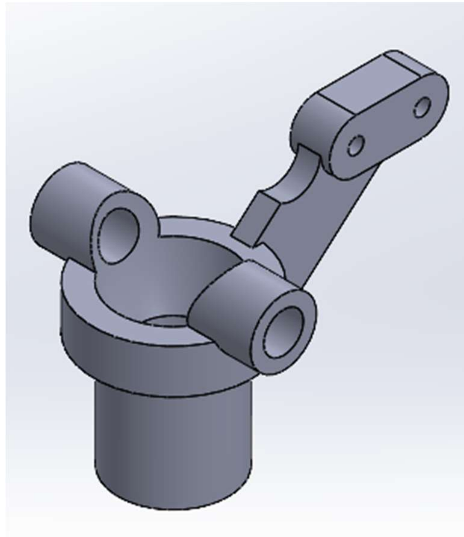

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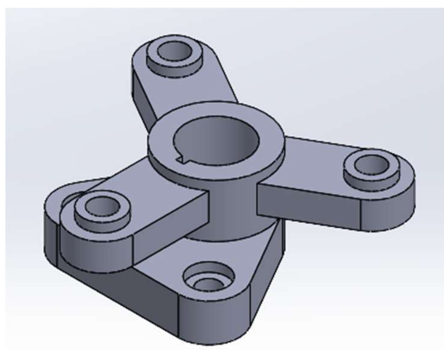

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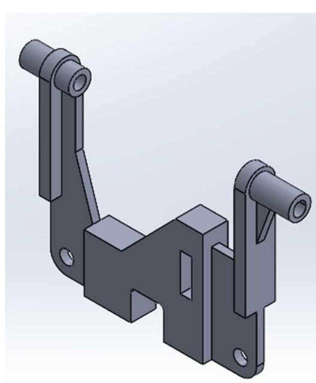

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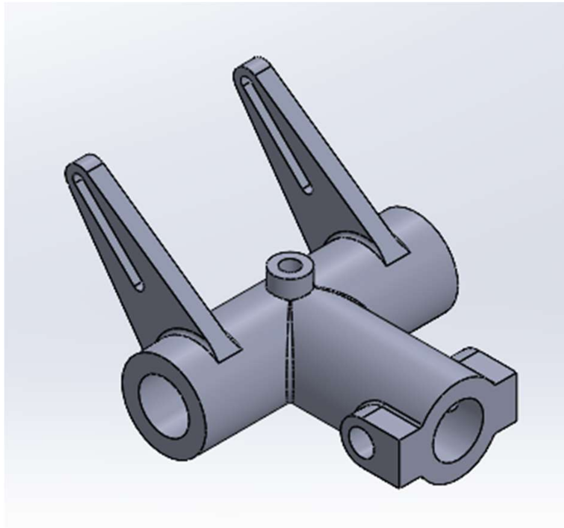

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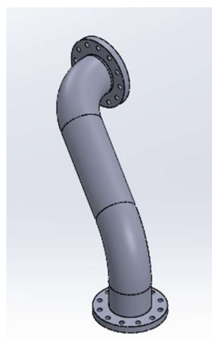

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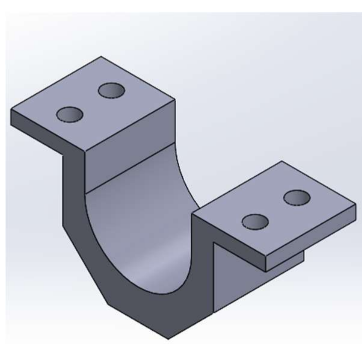
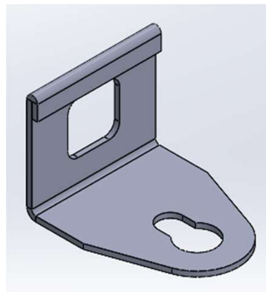


Sheet Metal Modelling

Assignment 1:



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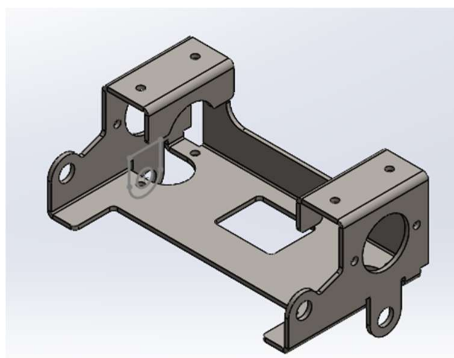

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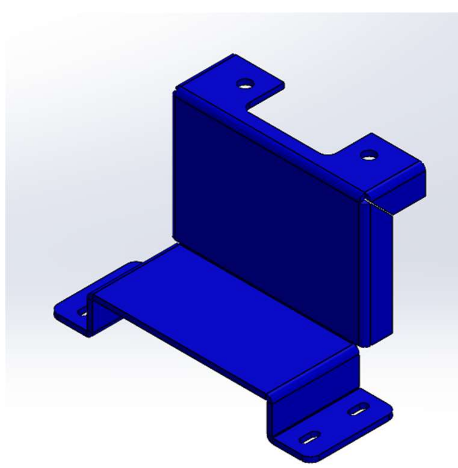

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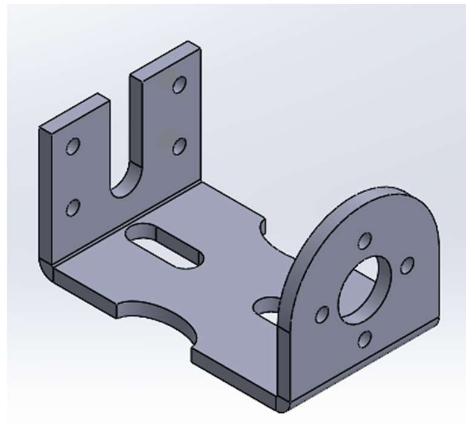
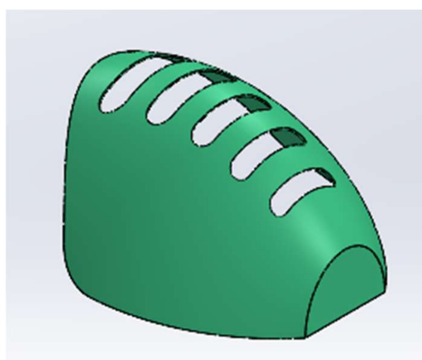


Surface Modelling

Assignment 1:



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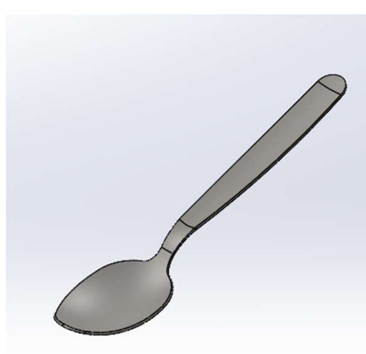

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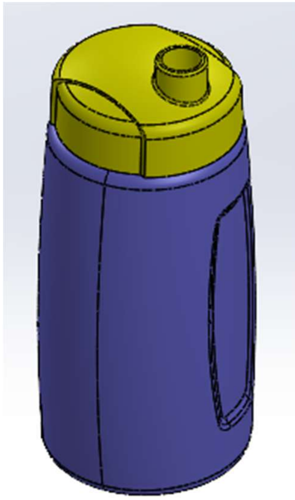
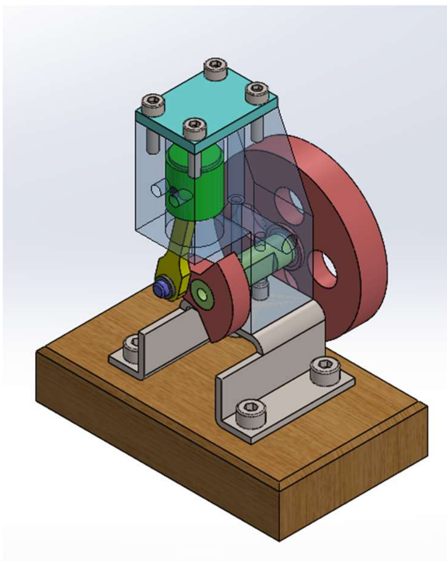
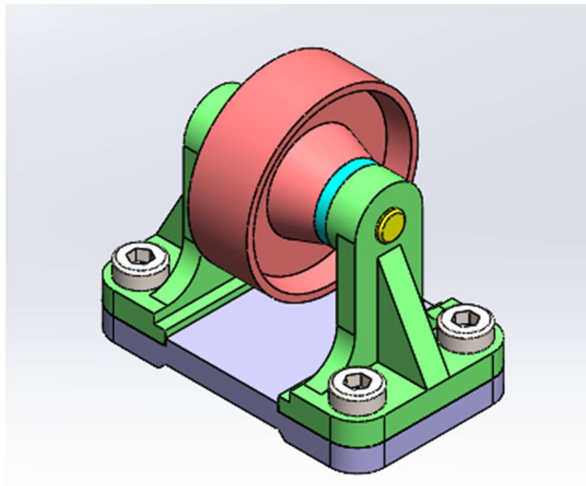


Assembly

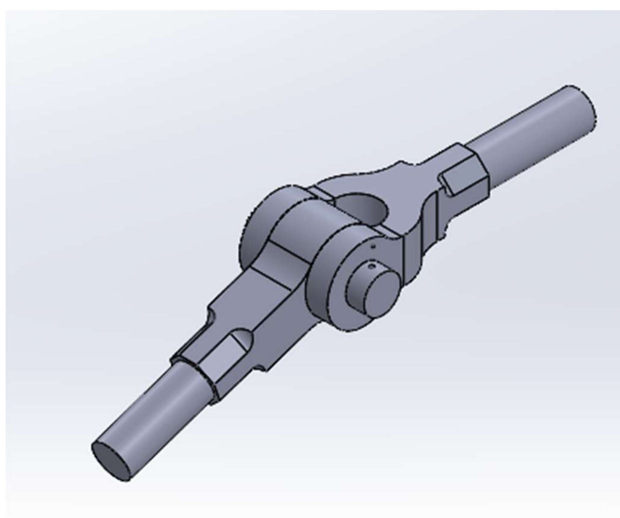
Assembly 1: Air Motor Assembly



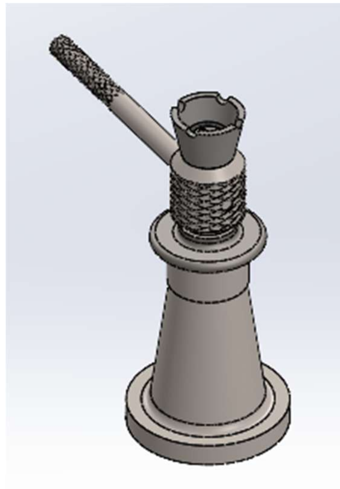
Assembly 2: Belt Roller Support



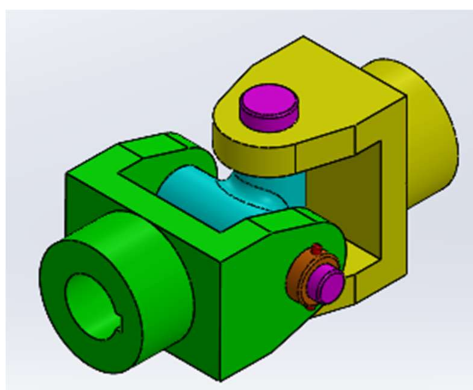
Assembly 3: Knuckle Joint



Assembly 4: Screw Jack

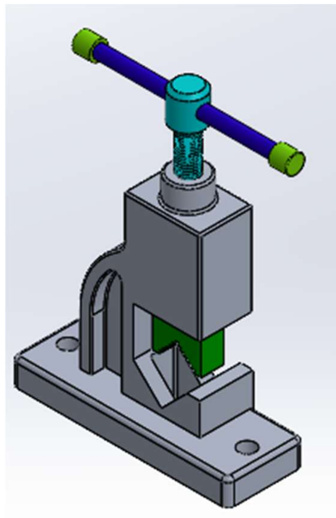


Assembly 5: Universal Joint

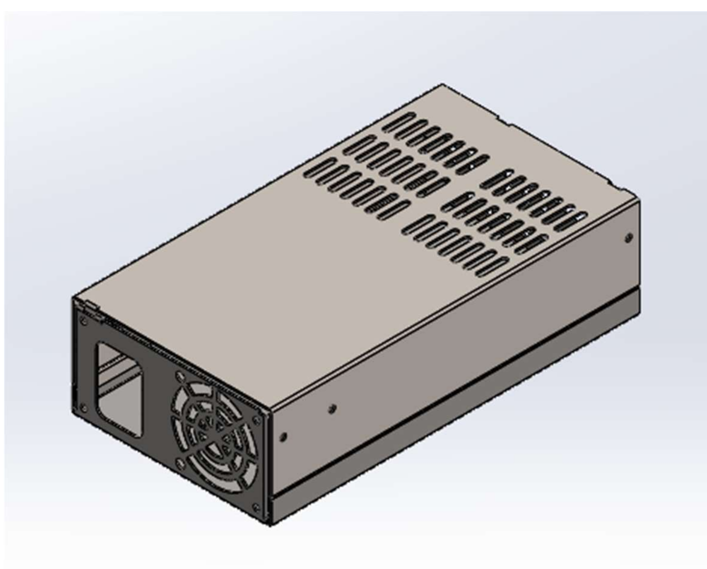


Mini project

Project 1: Pipe Vice Assembly



Project 2: Server Rack



Conclusion Remarks:

The four-month internship in AutoCAD and SolidWorks provided comprehensive technical training in computer-aided design and engineering modeling. During this period, I developed practical proficiency in 2D drafting, parametric 3D part modeling, assembly creation, and detailed engineering drawings. I gained hands-on experience in applying design constraints, dimensions, and geometric relationships to create accurate and fully defined models.

The training also enhanced my understanding of assembly relationships, motion considerations, and the preparation of manufacturing-ready drawings, including tolerances and standard annotations. Additionally, I worked with features such as sheet metal design, exploded views, and basic rendering, which strengthened my ability to visualize and communicate engineering concepts effectively.

This internship significantly improved my technical design skills, familiarity with industry-standard CAD workflows, and ability to translate conceptual ideas into precise digital models. Overall, the experience has strengthened my readiness to apply CAD tools in practical engineering applications and professional design environments.